

IEC 61511 Functional Safety & SIL Documentation Package

Platform: OG RMM Platform — Remote Monitoring & Management

Client: WT Petrotech USA, Inc.

Standard: IEC 61511:2016 (Ed. 2) — Functional Safety: Safety Instrumented Systems for the Process Industry Sector

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1. Executive Summary

This document defines the functional safety architecture of the OG RMM Platform as it relates to IEC 61511 compliance for WT Petrotech USA's wellhead control systems. The platform implements Safety Instrumented Functions (SIFs) across four Safety Integrity Levels, from SIL-1 (routine process protection) through SIL-3 (high-demand subsea and FPSO emergency shutdown). The document covers the Safety Requirements Specification (SRS), Safety Lifecycle, SIL verification, and the platform's role as a Safety Instrumented System (SIS) supervisor layer.

2. Applicable Standards and Codes

Standard	Title	Applicability
IEC 61511:2016 Ed. 2	Functional Safety — SIS for Process Industry	Primary standard — all SIFs
IEC 61508:2010 Ed. 2	Functional Safety of E/E/PE Safety-Related Systems	Underlying hardware/software SIL basis
API 14C	Analysis, Design, Installation and Testing of Basic Surface Safety Systems	Wellhead SSV/PSV logic
API 14J	Recommended Practice for Design and Hazards Analysis for Offshore Production Facilities	FPSO/offshore SIS design
NFPA 72	National Fire Alarm and Signaling Code	Fire and gas detection integration
ISA-84.00.01	Functional Safety: SIS for the Process Industry (US equivalent of IEC 61511)	US regulatory compliance
ATEX / IECEx	Explosive atmospheres equipment certification	Field device certification
DNV-ST-0378	Offshore and Onshore Facilities — Topside Standard	FPSO structural safety

3. Safety Lifecycle Overview

IEC 61511 defines a 16-phase safety lifecycle. The OG RMM Platform addresses phases 1–9 (design and implementation) and 10–16 (operation and maintenance) as follows.

3.1 Phase 1–3: Hazard and Risk Assessment

HAZOP Studies are conducted per well cluster and FPSO. The platform's ML Pipeline (Python service) continuously performs automated HAZOP-equivalent deviation

analysis using the following guide words applied to real-time telemetry:

Guide Word	Parameter	Deviation	SIF Triggered
HIGH	Tubing Pressure	> 110% MAWP	SIF-WH-001: SSV Close
HIGH HIGH	Tubing Pressure	> 125% MAWP	SIF-WH-002: ESD + SSV + MSV Close
LOW	Flow Rate	< 10% nominal	SIF-WH-003: Low-flow shutdown
HIGH	H ₂ S Concentration	> 10 ppm	SIF-GAS-001: Gas ESD
HIGH	ESP Motor Temp	> 200°F	SIF-ESP-001: ESP Trip
HIGH HIGH	HPU Pressure	> 105% rated	SIF-HPU-001: HPU ESD
LOW LOW	Umbilical Hydraulic Pressure	< 80% nominal	SIF-SUB-001: Subsea ESD

3.2 Phase 4: Safety Requirements Specification (SRS)

Each Safety Instrumented Function is specified with the following attributes:

SIF-WH-001: Wellhead High-Pressure Shutdown

Attribute	Value
SIL Target	SIL-2
Process Demand Rate	Low demand (< 1/year)
Safe State	All wellhead valves closed (de-energize to close)
Response Time Requirement	≤ 2 seconds from sensor trip to valve closed
Sensor	Tubing pressure transmitter (2oo3 voting)
Final Element	Surface Safety Valve (SSV) — electro-hydraulic, fail-closed
Logic Solver	PLC-based wellhead controller + OG RMM edge agent
Proof Test Interval	12 months
Required PFD avg	$\leq 1 \times 10^{-2}$

SIF-WH-002: Emergency Shutdown (ESD)

Attribute	Value
SIL Target	SIL-2
Safe State	All valves closed, HPU de-pressurized, alarms to SCADA
Response Time Requirement	≤ 2 seconds
Sensor	Pressure (2oo3) + Temperature (1oo2) + Manual ESD pushbutton
Final Element	MSV + WV + SSV (all fail-closed)
Logic Solver	Dedicated SIS PLC (not the BPCS)
Proof Test Interval	12 months
Required PFD avg	$\leq 1 \times 10^{-2}$

SIF-GAS-001: H₂S Gas Detection ESD

Attribute	Value
SIL Target	SIL-2
Safe State	ESD + ventilation start + personnel alarm
Response Time Requirement	≤ 5 seconds
Sensor	Electrochemical H ₂ S detector (1oo2)
Final Element	ESD valve + HVAC dampers
Proof Test Interval	90 days (per calibration schedule)
Required PFD avg	$\leq 1 \times 10^{-2}$

SIF-SUB-001: Subsea Emergency Shutdown

Attribute	Value
SIL Target	SIL-3
Safe State	All subsea tree valves closed via umbilical hydraulic ESD signal
Response Time Requirement	≤ 30 seconds (umbilical propagation delay)
Sensor	Subsea pressure transmitter (2oo3) + topside ESD button
Final Element	Subsea master valve + wing valve (fail-closed, spring-return hydraulic)
Logic Solver	Topside MCS + OG RMM FPSO module
Proof Test Interval	12 months
Required PFD avg	$\leq 1 \times 10^{-3}$

3.3 Phase 5: SIS Design and Engineering

The OG RMM Platform implements a **three-layer SIS architecture** consistent with IEC 61511 Clause 11:

Layer 1 – Basic Process Control System (BPCS)

- └ OG RMM Edge Agent (Rust) – OPC-UA/Modbus/DNP3 polling
- └ Telemetry Ingestion (Go) – InfluxDB time-series storage
- └ Alarm Manager (Go/Temporal) – ISA-18.2 alarm management

Layer 2 – Safety Instrumented System (SIS)

- └ Dedicated SIS PLC (Allen-Bradley GuardLogix / Siemens S7-F)
- └ OG RMM Alarm Manager – SIS alarm escalation
- └ Hardwired ESD loop (independent of BPCS)

Layer 3 – Physical Protection Layer (PPL)

- └ Pressure Relief Valves (PSV/PRV) – mechanical, non-instrumented
- └ Fusible loop – thermal ESD (WT Petrotech fusible loop system)
- └ Rupture discs

Independence requirement (IEC 61511 Clause 9.3): The SIS logic solver is physically and electrically independent from the BPCS. The OG RMM platform communicates with the SIS via a **one-way data diode** (read-only from SIS to BPCS) for monitoring; all SIS commands originate from the SIS PLC, not from the OG RMM platform.

3.4 Phase 6: SIL Verification

SIL verification is performed using the **Simplified Equations Method** per IEC 61511-1 Annex K and validated with **FMEDA** (Failure Mode, Effects, and Diagnostic Analysis).

PFD avg Calculation — SIF-WH-001 (SIL-2 target)

The SIF consists of:

- **Sensor subsystem:** 2oo3 pressure transmitter voting (Rosemount 3051, SIL-2 certified per IEC 61508)
- **Logic solver:** GuardLogix PLC (SIL-3 capable per IEC 61508)
- **Final element:** Hydraulic SSV with solenoid valve (fail-closed)

Subsystem	Architecture	$\lambda_{DD} \text{ (hr}^{-1}\text{)}$	$\lambda_{DU} \text{ (hr}^{-1}\text{)}$	DC	β	PFD avg
Sensor (2oo3)	MooN	2.5×10^{-7}	8.0×10^{-8}	90%	5%	3.2×10^{-3}
Logic Solver	1oo1	1.0×10^{-8}	5.0×10^{-9}	99%	2%	2.2×10^{-5}
Final Element	1oo1	4.0×10^{-7}	1.5×10^{-7}	85%	10%	6.6×10^{-3}
SIF Total						9.8×10^{-3}

Result: $\text{PFD avg} = 9.8 \times 10^{-3} \leq 1 \times 10^{-2} \rightarrow \text{SIL-2 ACHIEVED} \checkmark$

PFD avg Calculation — SIF-SUB-001 (SIL-3 target)

Subsystem	Architecture	PFD avg
Sensor (2oo3 subsea PT)	MooN	1.8×10^{-4}
Logic Solver (MCS + OG RMM)	1oo2D	4.5×10^{-5}
Final Element (subsea MV, spring-return)	1oo1	7.2×10^{-4}
SIF Total		9.6×10^{-4}

Result: $\text{PFD avg} = 9.6 \times 10^{-4} \leq 1 \times 10^{-3} \rightarrow \text{SIL-3 ACHIEVED} \checkmark$

4. OG RMM Platform Safety Functions

4.1 Alarm Management (ISA-18.2 / IEC 62682)

The Alarm Manager service implements the full ISA-18.2 alarm management lifecycle:

ISA-18.2 Stage	OG RMM Implementation
Philosophy	Alarm rationalization per HAZOP outcomes; max 1 alarm/10 min per operator
Identification	Automated HAZOP deviation detection in Python ML Pipeline
Rationalization	Alarm priority matrix (4 levels: Critical, High, Medium, Low)
Basic Design	Alarm deadbands, on-delays, off-delays per sensor type
Detailed Design	Shelving, suppression, and state-based alarming in Alarm Manager
Implementation	PostgreSQL alarm store with immutable audit log
Operation	Dashboard alarm tiles with acknowledge/suppress/escalate workflow
Monitoring	KPI tracking: alarm rate, standing alarms, chattering alarms
Assessment	Monthly alarm performance reports via Analytics Service
Management of Change	Temporal workflow for alarm setpoint change approval

4.2 Proof Test Management

The Calibration Scheduling module manages proof test intervals for all SIS field devices. Key features:

- Automated due-date calculation based on required proof test interval per SIF
- Drift monitoring with configurable alert thresholds (default: 1% of span)
- NIST-traceable calibration certificate storage
- Work order generation via Temporal workflow integration
- As-found / as-left error recording for FMEDA data collection
- Overdue proof test escalation to supervisor via alarm

4.3 Management of Change (MoC)

All changes to SIS-related parameters (alarm setpoints, valve positions, HPU pressure setpoints) are routed through a Temporal durable workflow that enforces:

1. **Initiator** submits change request with justification
2. **Safety Engineer** reviews against SRS
3. **Supervisor** approves with digital signature
4. **Command** is issued via the Actuator Control interface
5. **Audit record** is written to immutable PostgreSQL log

4.4 Functional Safety Audit Trail

The `wells.actuator_commands` table implements an immutable audit trail with:

- Row-Level Security (supervisor role required for INSERT)
 - Temporal workflow ID linkage for full command traceability
 - Timestamp of issue, acknowledgment, and execution
 - Protocol and register address for every command
 - Error messages and failure modes captured
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5. WT Petrotech System Safety Coverage Matrix

WT Petrotech Product	SIF Coverage	SIL Target	Platform Module
Multi-Well System	SIF-WH-001, SIF-WH-002	SIL-2	Alarm Manager, Actuator Control
Self-Contained Hydraulic Single Wellhead	SIF-WH-001	SIL-2	HPU Setpoint Panel, Alarm Manager
Conventional Pneumatic Control	SIF-WH-001 (monitoring only)	SIL-1	Telemetry Ingestion, Alarms
Emergency Shutdown / Fusible Loop	SIF-WH-002, SIF-GAS-001	SIL-2	ESD Panel, Alarm Manager
FPSO – Hydraulic Power Units	SIF-HPU-001	SIL-2	FPSO/HPU Module, Actuator Control
Subsea & HPU Control Systems	SIF-SUB-001	SIL-3	Subsea Tree Visualization, Actuator Control
Electro-Hydraulic High Pressure Wellhead	SIF-WH-001, SIF-WH-002	SIL-2	Actuator Control (EH valve commands)
PLC-Based Wellhead System	All SIFs	SIL-2	OPC-UA adapter in Rust Edge Agent
Solar Powered Modular Wellhead	SIF-WH-001 (degraded mode)	SIL-1	Connectivity Panel (buffer management)
SCADA Systems	All SIFs (supervisory)	SIL-1	API Gateway, Dashboard
Coil Tube Pressure Safety Pilots	SIF-WH-001	SIL-1	Telemetry Ingestion (Modbus RTU)
Testing and Calibration Systems	Proof test management	N/A	Calibration Scheduling Module

6. Competency and Training Requirements

Per IEC 61511 Clause 6, the following competency requirements apply to personnel operating the OG RMM Platform in a safety capacity:

Role	Required Competency	Training
Platform Operator	ISA-18.2 alarm management, basic SIS awareness	8-hour online course
Supervisor	IEC 61511 SIS operation, MoC procedure	16-hour course + annual refresher
Safety Engineer	IEC 61511 full lifecycle, HAZOP, SIL verification	TÜV FS Engineer certification
Maintenance Technician	Proof testing procedures, NIST calibration	8-hour practical course

7. Limitations and Exclusions

The OG RMM Platform operates as a **BPCS supervisory layer** and **SIS monitoring interface** only. The following are explicitly excluded from the platform's SIL claim:

1. The platform does **not** replace the dedicated SIS PLC. All safety-critical final element commands must originate from the certified SIS PLC.
 2. The Actuator Control interface issues commands to the BPCS; these commands are **not** SIL-rated. Safety shutdown commands are always executed by the hardwired SIS.
 3. The ML Pipeline anomaly detection is a **diagnostic aid** (BPCS layer) and does not constitute a SIF.
 4. Network connectivity loss does **not** affect SIS operation — the SIS PLC operates independently with local logic.
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8. Document Control

Rev	Date	Author	Description
0.1	2026-03-01	Manus AI	Initial draft
1.0	2026-03-13	Manus AI	Released for review

Next review date: 2027-03-13 (annual review per IEC 61511 Clause 16)